# A/B Test Experiment Summary & Recommendation

## Project: \"Quick Start Bundle\" Feature Evaluation

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# 1. Executive Summary & Decision

**Final Decision:** REJECT THE ROLLOUT & HALT THE TEST
While top-line experiment data displays a tempting **+7.87% lift in D7 ARPU**, a deep technical audit reveals that the experiment is fundamentally compromised. The test results cannot be used for business deployment due to two critical operational flaws: a severe **Sample Ratio Mismatch (SRM)** ($p \approx 8.23 \times 10^{-12}$) and an underlying **Android tracking/logging telemetry bug**. Moving forward with a rollout based on flawed assignment parameters poses a massive operational risk.

# 2. Core Experiment Readout & Methodology

### Handling Incomplete Cohorts

The core evaluation metrics rely heavily on 7-day maturation windows (D7 Retention, D7 ARPU). Because the experiment ran until February 25th, users acquiring the app near the tail end did not have a fully observed 7-day lifecycle. To prevent downward mathematical bias caused by incomplete data, all cohorts arriving after **2026-02-18** were isolated and excluded from D7 metrics.

- **Total Raw User Dataset:** 25,000 users
- **Mature D7 Cohorts (Analyzed):** 18,081 users

### Funnel Conversion Summary

The feature successfully accelerated top-of-funnel activity, prompting statistically significant improvements in onboarding checkpoints. However, this optimization failed to translate into core monetization, resulting in lower total payer acquisition.

| Metric Checkpoint | Control Group | Test Group | Lift (%) | P-Value | Statistical Significance |
|---|---|---|---|---|---|
| Tutorial Completion Rate | 71.86% | 73.37% | +2.11% | 0.0080 | Significant ($p < 0.05$) |
| Level 3 Completion | 46.99% | 48.84% | +3.93% | 0.0035 | Significant ($p < 0.05$) |

| Metric Checkpoint | Control Group | Test Group | Lift (%) | P-Value | Statistical Significance |
|---|---|---|---|---|---|
| Rate | | | | | |
| Day 1 Retention | 48.80% | 49.17% | +0.76% | 0.5553 | Not Significant |
| Day 7 Retention | 24.73% | 25.13% | +1.61% | 0.5375 | Not Significant |
| D7 Payer Rate | 2.30% | 2.24% | -2.34% | 0.8087 | Not Significant |

# 3. Revenue Stream Breakdown

Isolating the revenue components reveals that total monetization growth is an artifact of forced ad engagement rather than systemic core buying behavior. The bundle failed to safely scale premium financial conversions.

- **Ad Revenue ARPU (+10.63% Lift | p = 0.0000):** High statistical significance. The onboarding workflow aggressively integrated or exposed ad systems, multiplying impression volume per user early on.
- **In-App Purchase (IAP) ARPU (+7.03% Lift | p = 0.5456):** Statistically directionless. The feature failed to unlock robust premium transactional behavior.
- **Total Revenue ARPU (+7.87% Lift | p = 0.3777):** Statistically neutral. Because the core user sample is structurally unbalanced, this variance remains inside regular probability limits and cannot be safely scaled up.

# 4. Market Segmentation Deep-Dive

Reviewing user behavior across core system segments highlights a severe divergence between platform types and hardware configurations.

| Segment Identifier | Control D7 ARPU | Test D7 ARPU | Net Segment Lift (%) |
|---|---|---|---|
| **Platform: Android** | $0.2500 | $0.2340 | -6.41% |
| **Platform: iOS** | $0.2814 | $0.3497 | +24.25% |
| **Device Tier: High** | $0.2944 | $0.3084 | +4.74% |
| **Device Tier: Low** | $0.2727 | $0.2415 | -11.45% |
| **Device Tier: Mid** | $0.2501 | $0.2966 | +18.58% |

# 5. Data Quality, Anomalies & Sanity Checks

- **Fatal Sample Ratio Mismatch (SRM):** The assignment architecture systematically misallocated cohorts (Control: 8,581 users vs. Test: 9,500 users). A Chi-Square variance test confirms a systemic breakdown in user traffic assignment ($p = 8.23 \times 10^{-12}$). This mathematically proves that allocation was not random, invalidating top-line results.

- **Android Telemetry Bug:** In-depth historical slice analysis reveals that on February 10th and 11th, **100% of Android Test variant users possess null/missing values for tutorial_complete**, while the corresponding Control group registered data normally. The feature deployment likely induced application crashes or completely froze analytics logging frameworks across Android configurations during those dates.
- **Severe Platform Polarization:** The massive degradation on Android (-6.41% ARPU) contrasted against a surging iOS footprint (+24.25% ARPU) confirms that the feature's visual layout, monetization interface, or system optimization is deeply flawed on specific mobile operating environments.

# 6. Prescribed Next Steps

1. **Code Audit via Engineering:** Deliver tracking diagnostics to QA and Development teams to isolate and correct the event-logging failure observed on Android across the Feb 10–11 window.
2. **Re-balance the Randomization Layer:** Audit and patch the internal client/server gateway assigning buckets to stop assignment skewing and eradicate SRM flaws.
3. **Relaunch Stratified Test Framework:** Once systemic platform bugs are repaired, launch a clean \"Phase 2\" experiment. Treat iOS and Android as independent, completely isolated experimental swimlanes.